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Ryerson University
Department of Mechanical and Industrial Engineering

MTL 200 – Materials Science Fundamentals

Midterm Test No.1

February 18, 2011

Instructor: Professor Daolun Chen, *Dr.rer.nat., Ph.D., P.Eng.*

INSTRUCTIONS

1. Duration is **50 minutes.**
2. This is a closed book test.
3. **Questions 1 and 2 are compulsory. Answer any two more questions from #3 to #5, and tick the answered questions at the top of next page.**
4. Back pages may be used if extra space is needed.
5. Please use an ink pen/ballpoint pen, and do not use a pencil (except for drawings if any).
6. Total marks: **40**

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LAST NAME: [redacted] **FIRST NAME:** [redacted]
CAPITAL LETTERS CAPITAL LETTERS

STUDENT NO. [redacted]

SECTION NO. 9

Questions	1	2	3	4	5	Total
Questions answered	√	√	√	√		
Maximum marks	10	10	10	10	10	40
Actual marks	7	6	10	10		33

MARKS: 10 3

1. Indicate the correct answer by circling TRUE or FALSE.

(1) Protons and electrons have approximately the same mass.

TRUE / FALSE ✓

(2) In the wave-mechanical model, the electron is considered to exhibit only wavelike characteristics.

TRUE / FALSE ✗

(3) The smaller the principal quantum number, the lower the energy level.

TRUE / FALSE ✓

(4) Stable electron configurations mean that the states within the outermost shell are completely filled.

TRUE / FALSE ✓

(5) Valence electrons are the electrons in the outermost shell, which participate in the bonding between atoms.

TRUE / FALSE ✓

(6) In the periodic table, all elements that are arrayed in a given column or group have similar valence electron structures, and chemical and physical properties.

TRUE / FALSE ✓

(7) The covalent bonding is directional.

TRUE / FALSE ✗

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(8) There are four atoms in the body-centered cubic (BCC) unit cell.

TRUE / FALSE

(9) In the section of crystallographic directions and planes, the angle brackets < > indicate a family of equivalent planes.

TRUE / FALSE

(10) For an edge dislocation, the dislocation line and Burgers vector are parallel.

TRUE / FALSE

MARKS: 10 4

2. Circle the correct answer of the following questions.

(1) The three ages in the history of human civilizations which have been designated by the level of the materials development are

(a) Gold age, Silver age, and Bronze age.

(b) Gold age, Silver age, and Iron age.

(c) Stone age, Bronze age, and Ice age.

(d) Stone age, Bronze age, and Iron age.

(2) Three basic classifications of materials are

(a) metals, ceramics, and polymers.

(b) smart materials, ceramics, and polymers.

(c) metals, smart materials, and polymers.

(d) metals, ceramics, and smart materials.

(3) The number of energy states for *p* subshell is

(a) 1.

(b) 3.

(c) 5.

(d) 7.

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(4) Which type of bonding is the secondary bonding?

(a) metallic bonding.
(b) ionic bonding.
(c) van der Waals bonding.
(d) covalent bonding.

(5) The coordination number in the unit cell of body-centered cubic (BCC) crystal structure is

(a) 4.
(b) 8.
(c) 12.
(d) 16.

(6) The stacking sequence of close-packed planes in the form of ABCABCABC is for

(a) FCC.
(b) BCC.
(c) HCP.
(d) none of the above crystal structures.

(7) Four main factors affecting the solubility of the substitutional solid solution are

(a) composition, crystal structure, electronegativity, valence.
~~(b) atomic~~ size, composition, electronegativity, valence.
~~(c) atomic size, crystal~~ structure, composition, valence.
(d) atomic size, crystal structure, electronegativity, valence.

(8) Linear defects or imperfections in the crystalline materials normally mean

(a) vacancies.
(b) dislocations.
(c) grain boundaries.
(d) twin boundaries.

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(9) Two types of main diffusion mechanisms are

(a) interdiffusion and impurity diffusion.
(b) vacancy diffusion and interstitial diffusion.
(c) self-diffusion and interdiffusion.
(d) self-diffusion and interstitial diffusion.

(10) The driving force in the steady-state diffusion is

(a) diffusion coefficient.
(b) diffusion flux.
(c) concentration gradient.
(d) temperature.

10

3. The metal rubidium (Rb) has a BCC crystal structure. If the angle of diffraction for the (321) set of planes occurs at 27.00° (first-order reflection) when monochromatic x-radiation having a wavelength of 0.0711 nm is used, calculate (a) the interplanar spacing for this set of planes, and (b) the atomic radius for the rubidium atom.

$n\lambda = 2d_{hkl}\sin\theta$

$\theta = 13.5$
$\lambda = 0.0711\,nm$
$n = 1$

a) $1(0.0711) = 2\left(\frac{a}{\sqrt{3^2+2^2+1^2}}\right)\sin 13.5$

$\frac{0.0711}{2(\sin 13.5)} = \frac{a}{\sqrt{14}}$

$0.152(\sqrt{14}) = a$

$a = 0.570\,nm$

b) $a = \frac{4R}{\sqrt{3}}$

$0.570 = \frac{4R}{\sqrt{3}}$

$\frac{0.570(\sqrt{3})}{4} = R$

$R = 0.247$

10

4. Determine the approximate density and atomic weight of a nickel-base superalloy (*Hastelloy C-276 with typical applications in the pollution control, chemical processing, waste treatment, marine engineering, pulp and paper production, etc., Advanced Materials & Processes, January 2011, Vol.169, No.1, pp.19-22*). This alloy has a main composition of 17.0 wt% **Mo**, 16.5 wt% **Cr**, 7.0 wt% **Fe**, 4.5 wt% **W**, and the remainder **Ni**. 55%

Atomic weight

$$A_{ave} = \frac{100}{\frac{C_{Mo}}{A_{Mo}} + \frac{C_{Cr}}{A_{Cr}} + \frac{C_{Fe}}{A_{Fe}} + \frac{C_{W}}{A_{W}} + \frac{C_{Ni}}{A_{Ni}}}$$

$$= \frac{100}{\frac{17}{95.94} + \frac{16.5}{52} + \frac{7}{55.85} + \frac{4.5}{183.84} + \frac{55}{58.69}}$$

$$= \frac{100}{1.58}$$

$$= \boxed{63.23 \text{ amu}}$$

Density

$$\rho = \frac{100}{\frac{17}{10.22} + \frac{16.5}{7.19} + \frac{7}{7.87} + \frac{4.5}{19.3} + \frac{55}{8.90}}$$

$$= \boxed{8.88 \text{ g/cm}^3}$$

5. The activation energy for the diffusion of carbon in chromium is 111,000 J/mol. Calculate the diffusion coefficient at 827°C, given that D at 1127°C is 6.25×10^{-11} m²/s.

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